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RESEARCH ARTICLE

Efficacy of modern combination fungicide molecules against sheath blight of rice

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Abstract

Sheath blight of rice is one of the most damaging diseases of rice worldwide causing serious yield losses. Several fungicide molecules with different mode of actions are available in the market. The present study was undertaken to find the best and most economical combination of fungicides in managing this disease. Hence, an experiment was conducted to evaluate the efficacy and economics of flusilazole 12.5% + carbendazim 25% SC, difenoconazole 11.4% + azoxystrobin 18.2% SC, tebuconazole 18.3% + azoxystrobin 11% SC, tricyclazole 18% + mancozeb 62% WP, zineb 68% + hexaconazole 4% WP, trifloxystrobin 25% + tebuconazole 50% WG, mancozeb 50% + carbendazim 25% WS and epoxiconazole 62.5 g/l + fluxapyroxad 62.5 g/l EC against sheath blight of rice under the field conditions. Results of the present study revealed that, out of the different fungicidal combinations, two foliar sprays of difenoconazole 11.4% + azoxystrobin 18.2% SC @ 1 ml/lit was found to be most economic that provided best incremental net benefit ratio of 6.17:1.00. It was followed by two foliar sprays of epoxiconazole 62.5 g/l + fluxapyroxad 62.5 g/l EC @ 1.5 ml/l, tebuconazole 18.3% w/w + azoxystrobin 11% SC, zineb 68% + hexaconazole 4% WP and trifloxystrobin 25% + tebuconazole 50% WG.

Keywords Sheath blight · *Rhizoctonia solani* · Fungicides · Rice

Introduction

Sheath blight caused by *Rhizoctonia solani* Kuhn is one of the most destructive diseases of rice throughout the world causing substantial quantitative losses and quality degradation (Singh et al. 2013; Laha et al. 2017). *Rhizoctonia solani* infect wide range of plants of about 32 families belonging to 188 genera (Gangopadyay and Chakrabarti 1982). Under favorable climatic conditions the damage can reach upto 100% and depending upon the disease severity the loss in yield of rice may vary from 5.2–50% (Kannaiyan and Prasad 1978, 1984; Rajan 1987). Growing resistant variety is one of the best options to manage the disease. Unfortunately, till date no resistant variety is available to the farmers. Hence, integrating available disease management practices including use of varieties with fare degree of resistance, practices that minimize initial inoculum and fungicidal spray, is a most effective and economic way to minimize the losses caused by this disease. The majority of the fungicides namely; carbendazim, mancozeb, carboxin, zineb, benomyl, thiophanate, chloroneb, edifenphos, captafol etc. have been reported to be effective against this disease under field conditions (Dash and Panda 1984; Kannaiyan and Prasad 1984; Singh and Sinha 2004). Farmers are applying 3–4 sprays of these fungicides for the management of this disease. Use of a single fungicide with one mode of action will always have chances of development of resistance against the pathogen. Several new molecules with different combinations such as azoxystrobin in combination with difenoconazole and tebuconazole, carbendazim with mancozeb and flusilazole, hexaconazole with zineb, propiconazole with tricyclazole, tebuconazole with trifloxystrobin have been registered in India for use against rice sheath blight and are available in the market. However, farmers are confused as to which combination is best for the management of this disease. Therefore, keeping this in view the present investigation was carried out to determine the best and most economical fungicides combination for the management of the disease under field conditions.

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Materials and methods

Field experiments were carried out to test the efficacy of new molecules against sheath blight of rice during kharif 2017 and 2018 crop seasons at Norman E. Borlaug Crop Research Centre of G.B. Pant University of Agriculture and Technology, Pantnagar. The experiments were laid out in a Randomized Block Design (RBD) with three replications and nine treatments in 5 × 5 m plots. The susceptible variety Pant Dhan 4 was transplanted on 20th June, 2017 and 25th June, 2018 with 20 cm row to row and 10 cm plant to plant spacing. The crop was raised as per the recommended agronomic practices.

For artificial inoculation of the crop, the pathogen was isolated from sheath blight infected plants in 2016 and was multiplied under laboratory conditions. The pure culture of *R. solani* was multiplied on autoclaved rice culm bits (5–7 cm) medium in autoclaved polypropylene bags and incubated at 25 ±2 °C for 15 days. The crop was inoculated artificially at tillering stage by placing the inoculum between the tillers just above the water line. Dates of inoculation were: 6th Sept, 2017 and 17th Aug, 2018. Two foliar sprays of fungicides were given at 15 days interval. First spray was given 10 days after inoculation, using a hand operated knapsack sprayer fitted with hollow cone nozzle and water volume of 500 l/ha was maintained. Fungicide applications were made on Sept 15th and Sept 30th during 2017 and Aug 27th and Sept 11th during 2018.

Disease assessment

The data on disease incidence and severity were recorded 15 days after last spray. From each treatment, 20 randomly affected hills were selected and the plants were assessed for vertical spread of the disease individually using SES scale (0–9 scale) (Anonymous 1996).

The grain yield was recorded on plot basis and converted to kg/ha. Per cent increase in grain yield was calculated using following formula:

$$\mathrm{AYL} = \left[(\mathrm{Yp} - \mathrm{Yu})/\mathrm{Yp}\right] \times 100$$

where, Yp = Yield under protected condition and Yu = Yield under unprotected condition (Kumar 2011, 2012; Nagaraja et al. 2012).

Economics of the treatments

Overall efficacy and economics of these treatments in managing sheath blight disease was worked out by mean incidence/severity, grain yield, additional net income and increment net benefit - cost (B:C) ratio. Minimum support price of paddy @ Rs. 1750/q of the year 2018–2019 was considered for the purpose. For spraying 500–750 L of water in one hectare area, five man days were considered. Labour and spray charges were taken into account to compute incremental net benefit cost ratio. Cost of labour @ 316 per manday and market price of Lusture 37.5 SE (flusilazole 12.5% + carbendazim 25% SC) (@ Rs. 3550/lit), Amistar Top 325 SC (difenoconazole 11.4% + azoxystrobin 18.2% SC) (@ Rs. 468/lit), Custodia (tebuconazole 18.3% + azoxystrobin 11% SC) (@ Rs. 3596/lit), Merger (tricyclazole 18% + mancozeb 62% WP) (@ Rs. 1300/lit), Avatar (zineb 68% + hexaconazole 4% WP) (@ Rs. 860/kg), Nativo 75 WG (trifloxystrobin 25% + tebuconazole 50% WG) (@ Rs. 7000/kg), Sprint (mancozeb 50% + carbendazim 25% WS) (@ Rs. 1000/kg), and Adexar (epoxiconazole 62.5 g/l + fluxapyroxad 62.5 g/l EC) (@ Rs. 2563/lit) were taken to assess the incremental cost benefit ratio for different combination of treatments used in the investigations.

Statistical analysis

The data collected were analyzed statistically using analysis of variance (ANOVA) for randomized block design with the help of statistical programme STPR-1 developed by Department of Mathematics and Statistics, College of Basic Sciences and Humanities, G.B. Pant University of Agriculture and Technology, Pantnagar. In case of significant result, critical difference at 5% level of probability was taken for comparison between treatments means (Gomez and Gomez 1984).

Results

Effect of fungicides on severity of disease

The sheath blight severity ranged from 40.22 to 68.25% in 2017 and 29.63 to 50.28% in 2018. It is evident from the results that all the treatments significantly reduced disease severity as compared to the untreated check (Table 1). The reductions in disease severity by these fungicide treatments ranged between 20.09 and 41.07%. However, foliar spray of difenoconazole 11.4% + azoxystrobin 18.2% SC @ 1.0 ml/l was found to be the most effective fungicide that provided 41.07% control of the disease. Nevertheless, it was significantly at par with epoxiconazole 62.5 g/l + fluxapyroxad 62.5 g/l EC and tebuconazole 18.3% w/w + azoxystrobin 11% SC which provided 40.64 and 37.24% control of the sheath blight, respectively.

Table 1 Effect of various fungicide combinations on the severity of sheath blight disease of rice

Treatments		Dose	Disease severity as PDI (%)			Disease control (%)	Grain Yield (Kg/ha)			Increase in yield (%)
			2017	2018	Mean		2017	2018	Mean	
T1	Flusilazole 12.5% + carbendazim 25% SC	1.0 ml/l	44.80c	36.35b	40.58c	31.53	5607.00b	5639.33b	5623.17c	3.34
T2	Difenoconazole 11.4% w/w + azoxystrobin 18.2% w/w SC	1.0 ml/l	40.22a	29.63a	34.93a	41.07	6089.67a	6072.00a	6080.83a	11.75
T3	Tebuconazole 18.3% w/w + azoxystrobin 11% SC	1.5 ml/l	41.34a	33.07a	37.20b	37.24	5986.33a	5934.00a	5960.17a	9.53
T4	Tricyclazole 18% + mancozeb 62% WP	2.5 g/l	48.80d	45.92d	47.36e	20.09	5502.33c	5505.67c	5504.00d	1.15
T5	Zineb 68% + hexaconazole 4% WP	2.5 g/l	42.53b	35.13b	38.83b	34.49	5727.33b	5819.67b	5773.50b	6.10
T6	Trifloxystrobin 25% + tebuconazole 50% WG	0.4 g/l	43.41b	36.28b	39.84c	32.78	5639.00b	5645.00b	5642.00c	3.68
T7	Mancozeb 50% + carbendazim 25% WS	2.5 g/l	45.92c	38.96c	42.44d	28.40	5523.00c	5559.33c	5541.17c	1.83
T8	Epoxiconazole 62.5 g/l + fluxapyroxad 62.5 g/l EC	1.5 ml/l	40.42a	29.93a	35.18a	40.64	5989.67a	5998.67a	5994.17a	10.16
T9	Control		68.25	50.28	59.27	–	5423.67c	5459.33c	5441.50d	0.00
	Sem		0.56	1.17	0.61		44.41	67.18	41.74	
	CD at 5%		1.69	3.55	1.84		134.72	203.77	126.60	

Effect of fungicides on rice grain yield and economics of the treatments

Data presented in Table 1 revealed that two foliar sprays of difenoconazole 11.4% + azoxystrobin 18.2% SC @ 1.0 ml/l was found to be the best among all the treatments. The treatment provided highest grain yield (6080.83 kg ha^{-1}) and highest increase (11.75%) in grain yield over control. However, it was significantly at par epoxiconazole 62.5 g/l + fluxapyroxad 62.5 g/l EC and tebuconazole 18.3% + azoxystrobin 11% SC provided 5994.17 and 5960.17 kg ha^{-1} grain yield and 10.16 and 9.53% increase in grain yield, respectively.

It is evident from the results presented in Table 2 that among different fungicidal treatments two foliar sprays of difenoconazole 11.4% + azoxystrobin 18.2% SC @ 1 ml/lit was found to be most economical that provided best incremental net benefit ratio of 6.17:1.00. It was followed by two foliar sprays of epoxiconazole 62.5 g/l + fluxapyroxad 62.5 g/l EC @ 1.5 ml/l, tebuconazole 18.3% w/w + azoxystrobin 11% SC, zineb 68% + hexaconazole 4% WP and trifloxystrobin 25% + tebuconazole 50% WG.

Discussion

In the present study, all the fungicidal treatments significantly reduced the disease as compared to the unsprayed control. Results from the present study indicate that azoxystrobin in combination with difenoconazole and tebuconazole and fluxapyroxad in combination with epoxiconazole are most effective in managing the sheath blight disease of rice. The treatments provided more than 35% control of the disease severity and more than 30% reduction in incidence of the disease.

The results of the current study are in conformity with previous studies that reported that different fungicide combinations namely; difenoconazole + azoxystrobin (Bhuvaneswari and Krishnam Raju 2012), epoxiconazole + fluxapyroxad (Prasanna and Veerabhadraswamy 2014), trifloxystrobin + tebuconazole (Bag 2009; Visalakshmi et al. 2016), tebuconazole + azoxystrobin (Jin et al. 2013), flusilazole + carbendazim (Bag et al. 2009), propiconazole + tricyclazole (Udhayakumar et al. 2019), tricyclazole + mancozeb (Chethana 2018), hexaconazole + zineb (Kumar et al. 2018), carbendazim and mancozeb (Thangasamy and Rangaswamy 1989) significantly reduced sheath blight and minimize the yield losses.

Of the eight fungicide combinations six fungicide combinations difenoconazole plus azoxystrobin, trifloxystrobin plus tebuconazole, tebuconazole plus azoxystrobin, flusilazole plus carbendazim, propiconazole plus tricyclazole, hexaconazole plus zineb, carbendazim and mancozeb have

Table 2 Economics of the fungicidal treatments in managing the disease and improving the grain yield of rice

Treatments		Dose	Mean Grain Yield (Kg/ha)	Additional yield (Kg/ha)	Additional cost (Rs/ha)	Additional return (Rs./ha)	Incremental net B:C ratio
T1	Flusilazole 12.5% + carbendazim 25% SC	1.0 ml/l	5623.17c	181.67	3355.00	3179.23	0.95:1.00
T2	Difenoconazole 11.4% w/w + azoxystrobin 18.2% w/w SC	1.0 ml/l	6080.83a	639.33	1814.00	11188.28	6.17:1.00
T3	Tebuconazole 18.3% w/w + azoxystrobin 11% SC	1.5 ml/l	5960.17a	518.67	4277.00	9076.73	2.12:1.00
T4	Tricyclazole 18% + mancozeb 62% WP	2.5 g/l	5504.00d	62.50	3205.00	1093.75	0.34:1.00
T5	Zineb 68% + hexaconazole 4% WP	2.5 g/l	5773.50b	332.00	3429.00	5810.00	1.69:1.00
T6	Trifloxystrobin 25% + tebuconazole 50% WG	0.4 g/l	5642.00c	200.50	2980.00	3508.75	1.18:1.00
T7	Mancozeb 50% + carbendazim 25% WS	2.5 g/l	5541.17c	99.67	2830.00	1744.23	0.62:1.00
T8	Epoxiconazole 62.5 g/l + fluxapyroxad 62.5 g/l EC	1.5 ml/l	5994.17a	552.67	3502.25	9671.73	2.76:1.00
T9	Control	–	5441.50d	–	–	–	–
	Sem		41.74				
	CD at 5%		126.60				

been registered in India for the use against sheath blight of rice, however, tricyclazole 18% + mancozeb 62% WP and epoxiconazole 62.5 g/l + fluxapyroxad 62.5 g/l EC have not been registered in India but their efficacy has been confirmed (Chethana 2018; Prasanna and Veerabhadraswamy 2014).

The present findings provide valuable information that will assist farmer in selecting the best and most economic fungicide combination having different modes of action for management of sheath blight disease of rice. Based on the overall efficacy and economics of the treatments two foliar sprays of difenoconazole 11.4% w/w + azoxystrobin 18.2% w/w SC @ 1 ml/lit may be advocated to the farmers for better management of sheath blight of rice and obtaining good yield and returns. Care should be taken that indiscriminate use of fungicides with one or same mode of action may lead to the risk of the development of resistance to the fungicide. Farmers can get good yield of paddy if they adopt combination approach (Kumar et al. 2016).

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